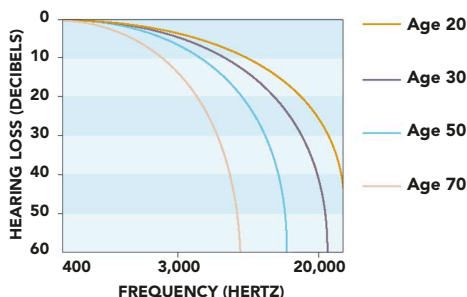


NERVOUS SYSTEM

As people age, the brain and nervous system undergo changes, losing nerve cells. Messages are transmitted more slowly, and the senses may be affected.

HEARING

More than half of people over 60 have hearing difficulties. Problems are caused by changes in the cochlea in the inner ear (see p.117). At birth, there are about 15,000 hair cells in the inner ear, but they gradually reduce with age, and the body is unable to generate new cells.



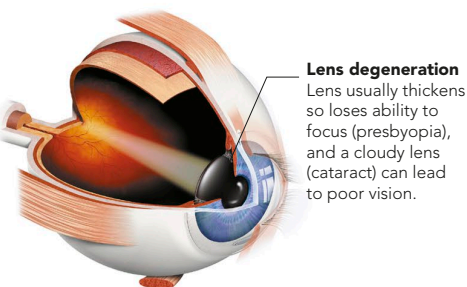
HIGH-FREQUENCY DROP-OFF

Ageing usually causes a loss of sensitivity to sounds; they may become dull or distorted so that speech is difficult to follow. The first sign is often difficulty hearing high-frequency sounds. Hearing aids may enhance the ability to understand speech.

Reflexes may be lost, leading to problems with movement and safety. Waste products may collect in the brain tissue. Some slowing of memory and thinking occurs.

VISION

Older people are susceptible to a number of visual disorders (see p.129). In a cataract, for example, the normally transparent lens of the eye becomes cloudy. Meanwhile, macular degeneration can affect the retina, causing detailed vision to deteriorate.

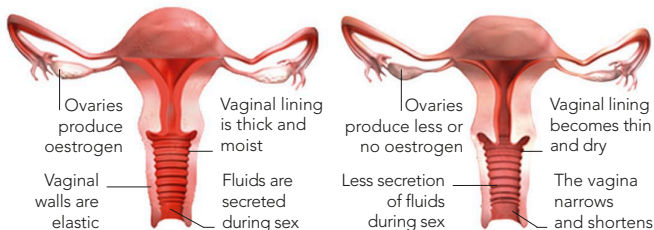


COMMON EYE PROBLEMS

A range of eye and visual disorders, particularly those involving the ability of the lens to focus, is more likely to occur with increasing age. Presbyopia, in which people lose the ability to adjust their eyes so they can see nearby objects, is almost universal.

THE MENOPAUSE

The menopause results from decreasing production of sex hormones. Symptoms include hot flushes, insomnia, night sweats, and headaches. Falling oestrogen levels can also cause depression. Menstruation may be irregular for several years up to the menopause, which is complete once a woman has not had a period for one year. The average age for the menopause in developed countries is 51 years.



PREMENOPAUSAL VAGINA

Before the menopause, the vaginal lining is thick and well lubricated; the walls stretch easily and mucous fluids are secreted.

POST MENOPAUSAL VAGINA

Declining oestrogen levels cause a reduction in vaginal mucus production; the vagina walls lose some elasticity and become thinner.